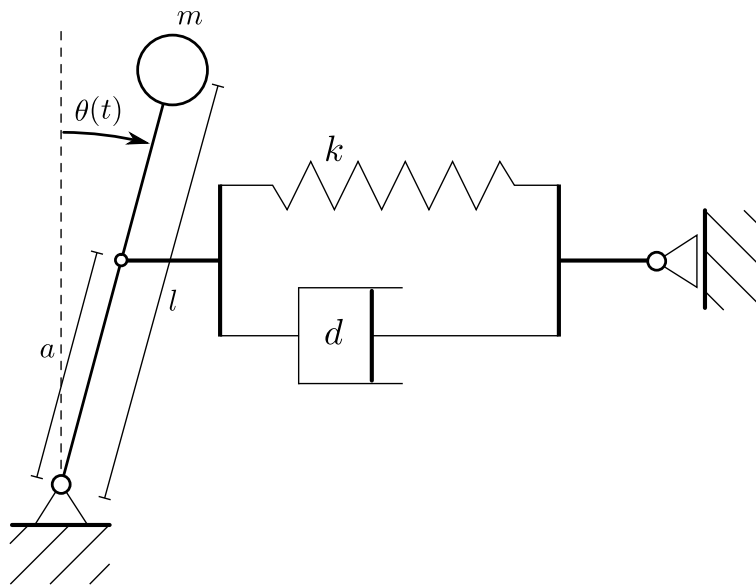


Solution 01
System Modeling, Linearization & Discretization

Exercise 1

We consider the following problem :



- The spring is at rest at $\theta(t) = 0$.
 - The resulting spring-damper force always acts horizontally on the pendulum.
 - We neglect the vertical movement of the spring-damper system.
1. Find the differential equation governing $\theta(t)$ for the autonomous non-linear pendulum system in state space form.
 2. Simulate the system from different initial positions $\theta(0)$, $\dot{\theta}(0)$.

Solution:

1. We use Newton's law to derive the equations of motion by summing up all torques T_i acting on the system. It holds that

$$\sum T_i = \ddot{\theta} m l^2,$$

where the system inertia $m l^2$ depends on the point mass m at the radius l . The acting torques are

- $T_g = F_g l \sin(\theta)$,
- $T_{sd} = -F_{sd} a \cos(\theta)$,

with the respective forces

- $F_g = mg$,
- $F_{sd} = k a \sin(\theta) + d a \frac{\partial \sin(\theta)}{\partial \theta} \frac{\partial \theta}{\partial t} = k a \sin(\theta) + d a \cos(\theta) \dot{\theta}$.

For a state-space description we introduce the state vector

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix},$$

and finally find the non-linear dynamics

$$\dot{x} = f(x) = \begin{bmatrix} \frac{g}{l} \sin(x_1) - \frac{a^2}{ml^2} \cos(x_1) x_2 \\ k \sin(x_1) + d \cos(x_1) x_2 \end{bmatrix}.$$

2. Matlab

Exercise 2

1. Find all equilibrium points and linearize the system at one where $\theta(t) = 0$.
2. Compare linearized and nonlinear system in simulation.

Solution:

1. Equilibrium points are given by all states x such that $\dot{x} = f(x) = 0$:

- $x_2 \stackrel{!}{=} 0$,
- $\sin(x_1) - k \frac{a^2}{mgl} \cos(x_1) \sin(x_1) \stackrel{!}{=} 0$.

One set of solutions is given by $\sin(x_1) = 0$, i.e.

$$x_1 \stackrel{!}{=} n\pi, \quad n \in \mathbb{N}.$$

Another set is given by the solutions of

$$k \frac{a^2}{mgl} \cos(x_1) \stackrel{!}{=} 1,$$

which depends of the choice of parameters and has a solution if $k \frac{a^2}{mgl} \geq 1$. Note the symmetry in the solution!

Linearizing the system around $\theta = \dot{\theta} = 0$ yields

$$A^c = \left. \frac{\partial f(x)}{\partial x} \right|_{x=[0,0]^T} = \begin{bmatrix} 0 & 1 \\ \frac{g}{l} - \frac{a^2}{ml^2} k & -\frac{a^2}{ml^2} d \end{bmatrix}$$

and

$$\dot{x}^c = A^c x^c.$$

(Just for your information:) The eigenvalues of A^c are

$$\frac{-\frac{a^2}{ml^2} d \pm \sqrt{(\frac{a^2}{ml^2} d)^2 + 4(\frac{g}{l} - \frac{a^2}{ml^2} k)}}{2}.$$

The real part is negative if $\frac{g}{l} < \frac{a^2}{ml^2} k$, which indicates about the local stability of the nonlinear system.

2. Matlab takeaways:

- Equilibrium at 0 is locally stable if $k \frac{a^2}{mgl} \geq 1$, the other equilibrium is unstable.
- Linearization adequate when close to the linearization point. (Also true for unstable system.)

Exercise 3

1. Discretize the nonlinear system using an Euler-Forward scheme.
2. Discretize the linearized system using an Euler-Forward scheme.
3. Discretize the linearized system using Exact discretization.
4. Compare the resulting system descriptions.

Solution:

1. Approximate the derivative as

$$\dot{x}^c(t) = f(x^c(t)) \approx \frac{x^c(t + T_s) - x^c(t)}{T_s},$$

and rearrange to

$$x^c(t + T_s) = x^c(t) + f(x^c(t))T_s.$$

Using the notation $x(k) := x^c(kT_s)$, we find the approximate model as

$$x(k+1) = x(k) + f(x(k))T_s.$$

2. Similarly, we have

$$\dot{x}^c(t) = A^c x^c(t) \approx \frac{x^c(t + T_s) - x^c(t)}{T_s},$$

and rearrange to

$$x^c(t + T_s) = x^c(t) + A^c x^c(t)T_s.$$

Using again the notation $x(k) := x^c(kT_s)$, we find the approximate model as

$$x(k+1) = \underbrace{(I + A^c T_s)}_{A_{\text{euler}}} x(k).$$

3. The first order linear ODE $\dot{x} = A^c x$ with initial value $x(0)$ has the following analytic solution : $x(t) = e^{A^c t} x(0)$.¹ We derive the exact discrete representation

$$x(k+1) = \underbrace{e^{A^c T_s}}_{A_{\text{exact}}} x(k),$$

with

$$e^{A^c T_s} = \sum_{i=0}^{\infty} \frac{(A^c)^i}{i!} T_s^i = \underbrace{I + A^c T_s}_{A_{\text{euler}}} + \frac{(A^c)^2}{2} T_s^2 + \dots$$

This discretization exactly matches the continuous time system at the sampling times.

4. Matlab

¹It is also possible to get an analytic solution with constant input over the sampling time (i.e. piecewise constant).